

Data, Signals, and Events Used in Emergency Braking Communication

The following table presents the data, signals, and events required for interaction and communication between trucks during an emergency braking event. It identifies the information exchanged, the communication messages, and the events that trigger system actions.

Type	Name	Description
Data	Truck ID	Unique identifier of the truck sending status
Data	Speed	Current vehicle speed used to determine stopping state
Data	Brake Status	Indicates whether braking is active or completed
Data	V2V Link Status	Shows whether V2V communication is available or unavailable
Signal	Emergency Brake Signal	Broadcast from lead truck triggering immediate braking
Signal	Braking Acknowledgement	Sent by follower truck confirming brake activation
Signal	All Stopped	Sent by Platoon Manager confirming all trucks have stopped
Signal	Standby Command	Broadcast from Platoon Manager to enter post-emergency standby
Event	Hazard Detected	Lead truck sensors detect obstacle requiring immediate full stop
Event	Signal Received	Follower truck receives emergency brake broadcast
Event	Braking Applied	Truck begins decelerating at maximum braking force
Event	Full Stop Confirmed	Platoon Manager confirms all trucks have reached speed zero
Event	Standby Entered	All trucks transition to standby mode after full platoon stop

Communication Protocol for Emergency Braking

The emergency braking scenario requires ultra-low latency communication between the lead truck and all follower trucks to prevent rear-end collisions. Based on the UPPAAL model, communication involves a broadcast emergency signal, braking acknowledgements, stop confirmation, and a standby command exchanged within strict timing constraints.

A suitable protocol for this scenario is 5G Cellular Vehicle-to-Everything (5G C-V2X) communication. 5G C-V2X supports direct Vehicle-to-Vehicle (V2V) communication for broadcasting safety-critical signals and Vehicle-to-Infrastructure (V2I) communication for interaction with the Platoon Manager. The protocol provides the sub-100 ms latency required to ensure all follower trucks receive and act on the emergency signal before a collision can occur.

Communication Message	Protocol Type
Emergency Brake Signal	5G C-V2X (V2V)
Braking Acknowledgement	5G C-V2X (V2V/V2I)
All Stopped Confirmation	5G C-V2X (V2I/V2V)
Standby Command	5G C-V2X (V2V)
Sensor Fallback Trigger	Onboard Sensor (local)